



Company Growth Analytics System

A Computational System Study Using Firmographic And Macroeconomic Data To Deliver Actionable Sector Forecasts.

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Engineering

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Next





BACKGROUND, PROBLEM & MOTIVATION

Background

Traditional economic analysis predominantly relies on aggregate macroeconomic indicators (e.g., GDP, inflation, unemployment), which can miss variation across individual firms (Autor et al., 2021; Smith & Lee, 2022). Such aggregate approaches overlook firm-level heterogeneity and structural sector dynamics that influence economic outcomes (Jones & Pérez, 2023; Wu, 2024).



Problem Statement

- Existing economic intelligence systems are:
 - Static and retrospective
 - Limited to descriptive analytics
 - Lacking predictive and interactive capabilities

Motivation

- Increasing availability of large-scale firmographic datasets
- Need for real-time, predictive, and data-driven decision support systems
- Machine learning enables pattern discovery and predictive inference in high-dimensional



LITERATURE REVIEW

Scope of Review

- Approximately 35 academic studies were reviewed
- 18 studies directly relevant to this research

Key Themes

- Machine learning applications in economic and financial prediction
- Firm-level (microeconomic) data analysis
- AI-driven decision support systems

Key Findings

- Machine learning models outperform traditional econometric approaches in high-dimensional contexts
- Firm heterogeneity significantly influences sector performance and economic outcomes

Research Gap

- Limited integration of:
 - Firmographic data
 - Predictive machine learning
 - Interactive web-based analytics platforms





OBJECTIVES

Main Objective

- To develop a robust, machine learning-powered web platform that analyzes and clusters key economic indicators, enabling accurate prediction of company growth and providing actionable economic intelligence for strategic decision-making.



Design

Implement

Predict

Specific Objectives

- To collect and preprocess large-scale firmographic datasets
- To develop predictive and clustering models for company analysis
- To design an interactive web-based analytics platform
- To deploy the system using scalable cloud infrastructure

Evaluation Criteria

- Predictive accuracy
- Model interpretability
- System usability and responsiveness

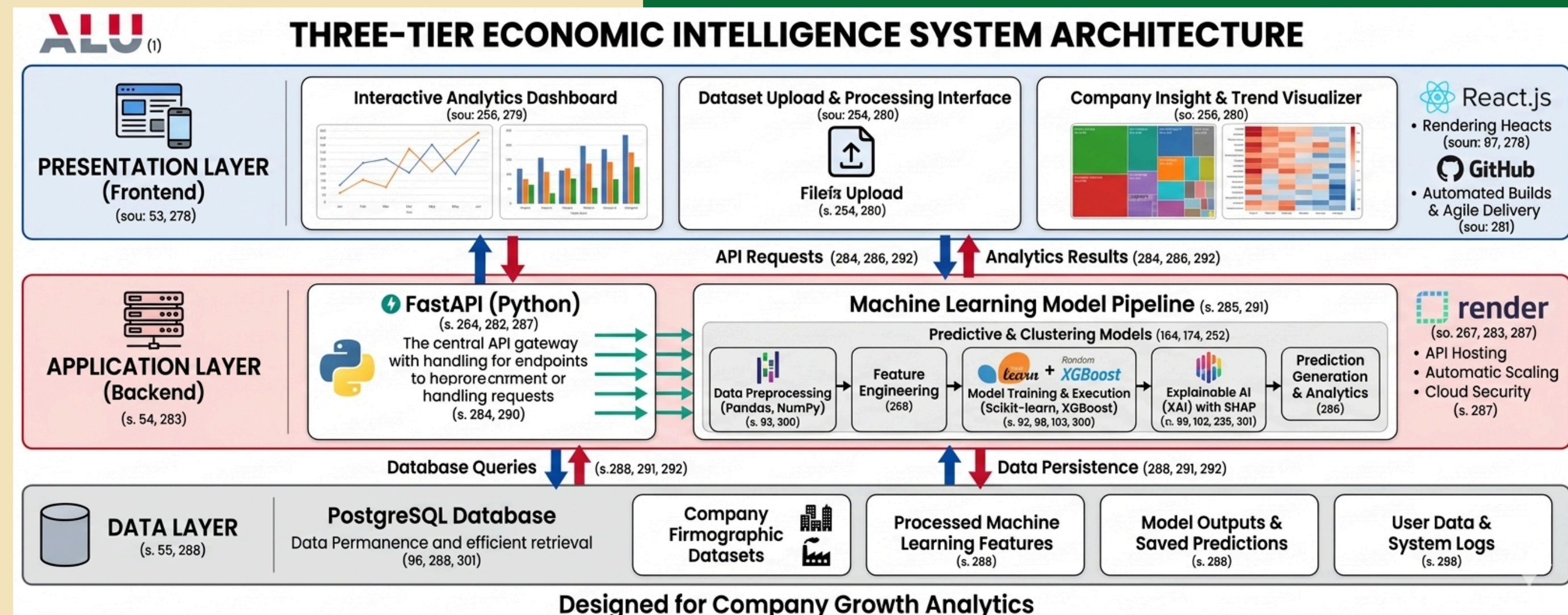
SYSTEM ARCHITECTURE

Architecture Type

- Three-tier web-based architecture
- System Components
- System Workflow

Design Considerations

- Scalability and modularity
- Real-time processing capability
- User-centered interactive design





DATA SOURCING & EXPLORATORY ANALYSIS



Data Characteristics

- 7million+ Large-scale firmographic company dataset
- High-dimensional feature space

Key Features

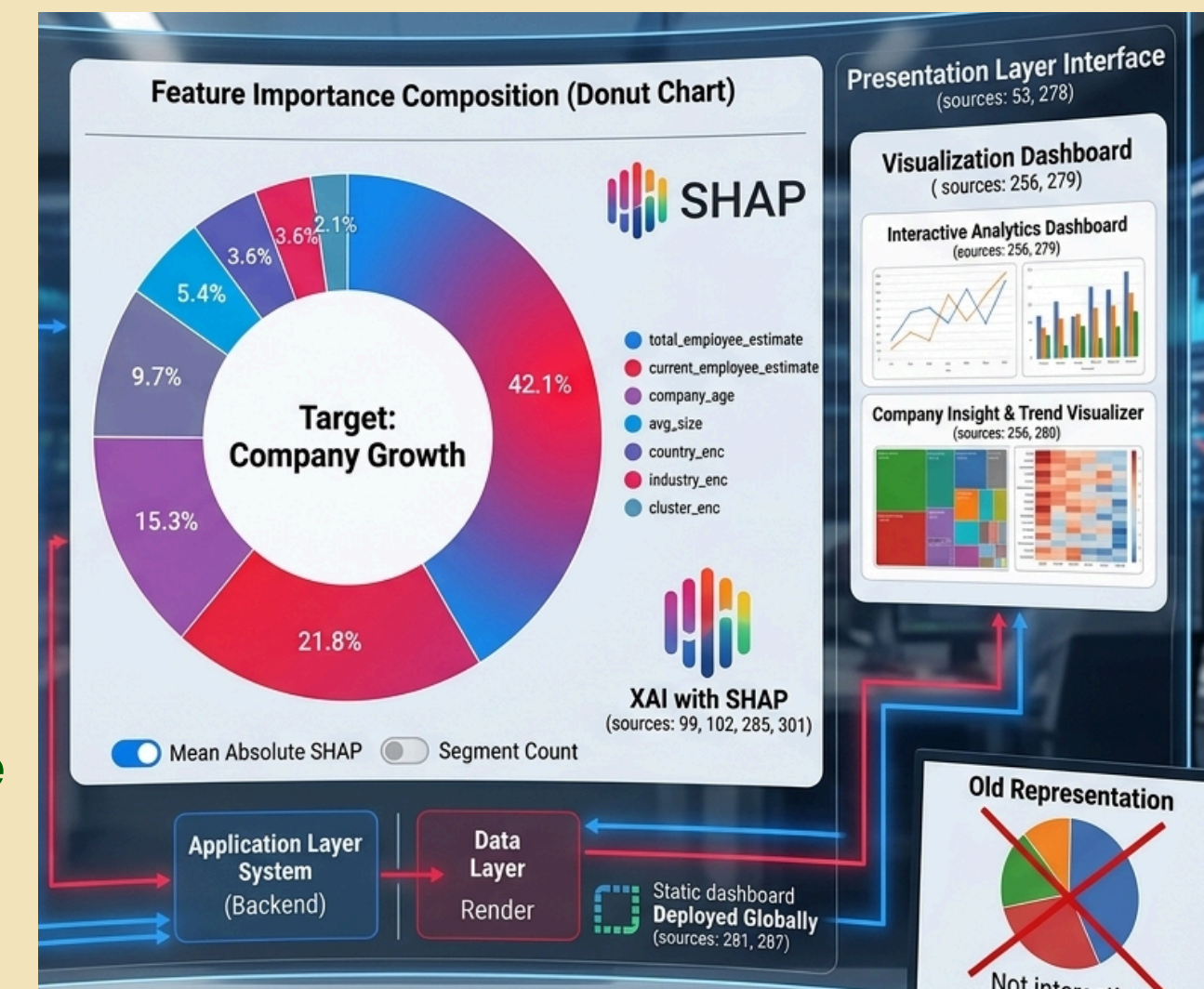
- Industry classification
- Company size and workforce metrics
- Geographic distribution
- Firm age

Exploratory Insights

- Presence of nonlinear relationships among variables
- Identification of latent clusters within company data
- Workforce-related features exhibit strong predictive relevance

Considerations

- Potential sampling bias
- Absence of detailed financial indicators



MODEL DESIGN & IMPLEMENTATION

Modeling Approach

- Supervised learning for prediction
- Unsupervised clustering for pattern discovery

Algorithms Implemented

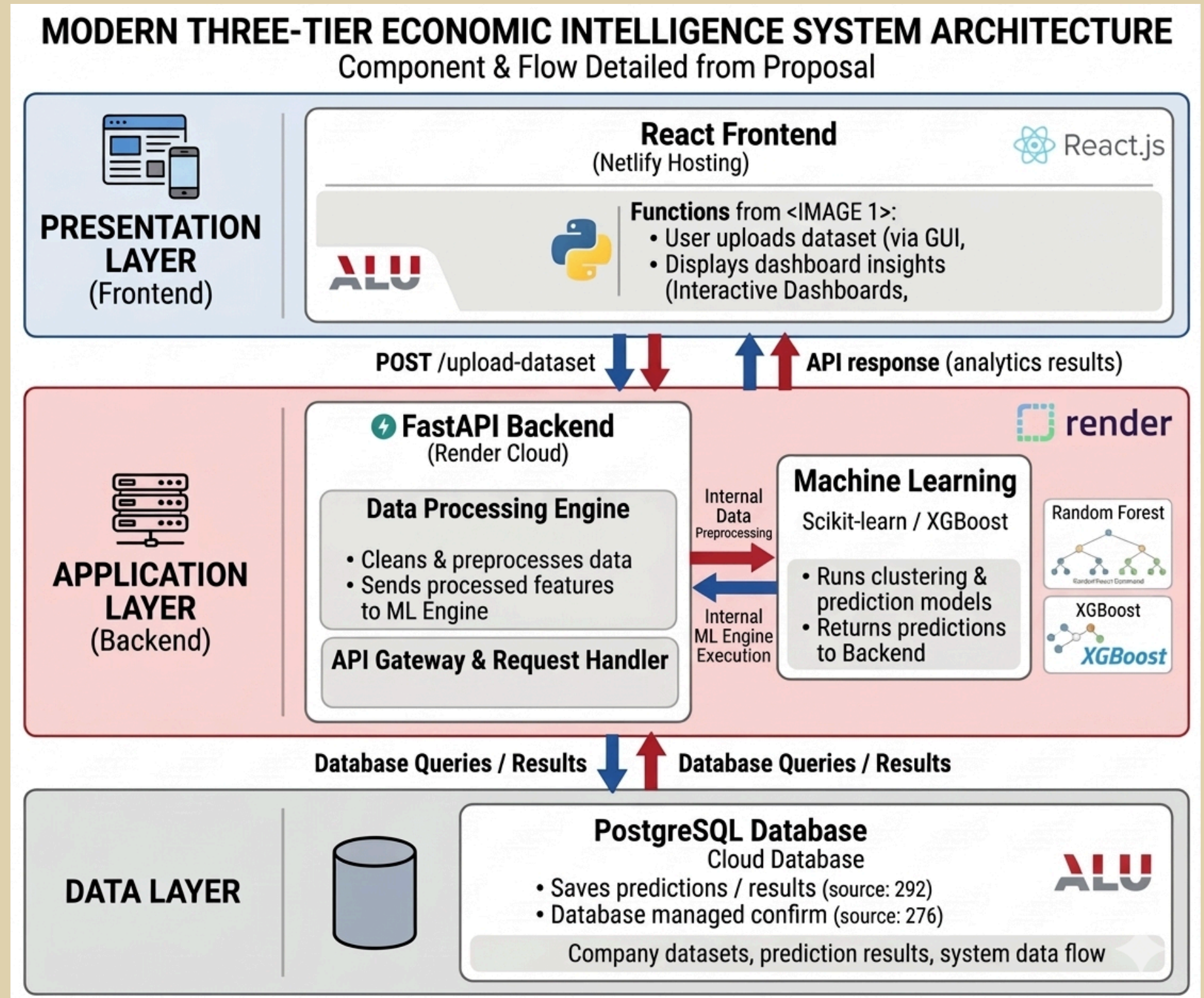
- Random Forest
- Extreme Gradient Boosting (XGBoost)

Technological Stack

- Python (FastAPI backend)
- scikit-learn, XGBoost
- React.js frontend

Core Functionalities

- Data preprocessing and feature engineering
- Model training and evaluation
- Prediction generation and clustering analysis





TRAINING, EVALUATION & RESULTS



Training Strategy

- Preprocessing and normalization of input data
- Model training on sampled datasets for computational efficiency

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score

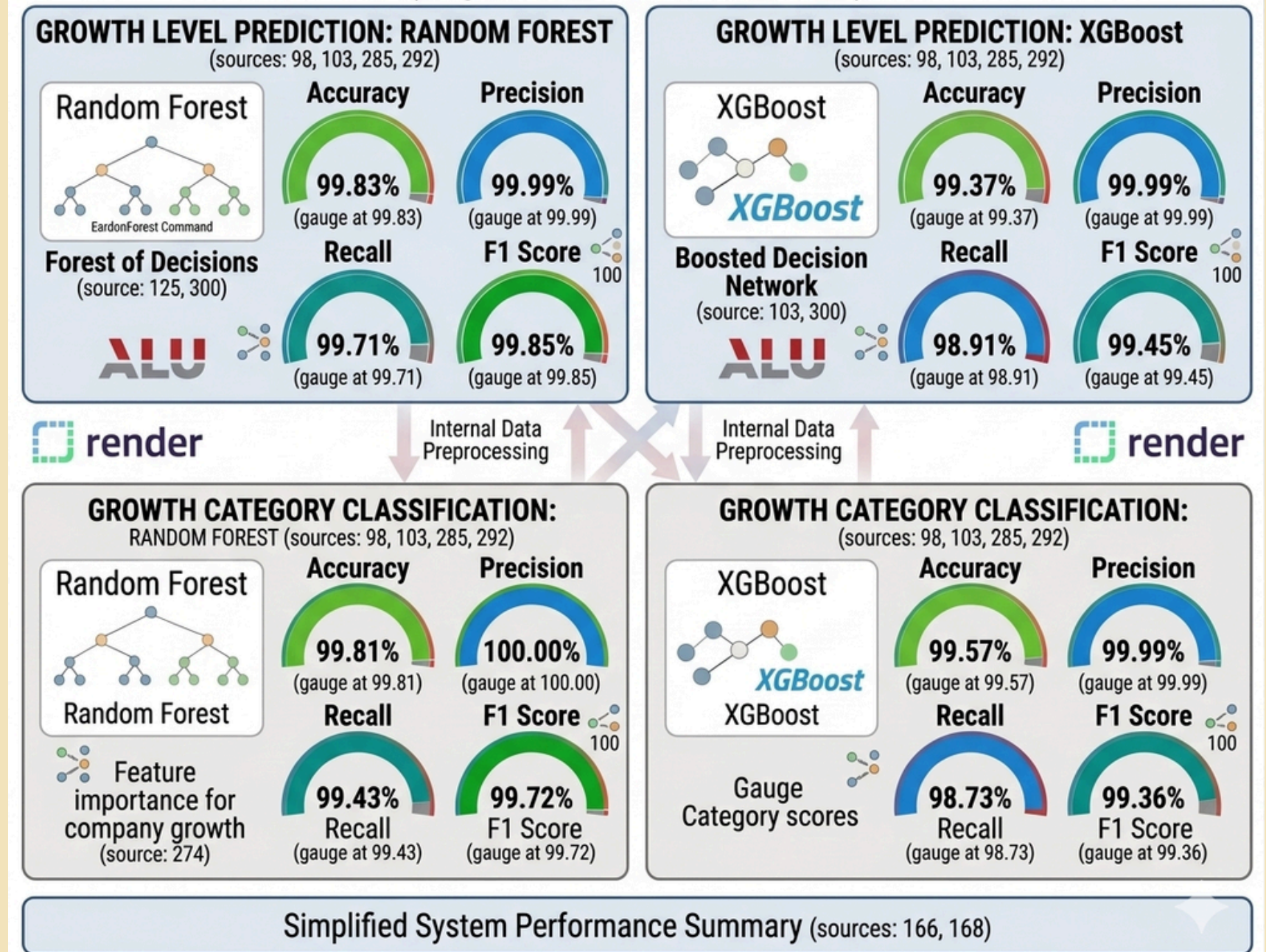
Results

- Both models achieved high predictive performance, with Random Forest slightly outperforming XGBoost

Interpretation

- Strong relationships exist between firmographic attributes and growth outcomes
- High accuracy suggests effective pattern learning,

SIMPLE PERFORMANCE METRICS GRAPH: ML MODEL GROWTH ANALYSIS (targets from source: 164, 256)



SYSTEM PERFORMANCE & APPLICATION

System Capabilities

- Real-time prediction and analytics
- Interactive visualization dashboards
- Scalable cloud-based deployment

Performance Indicators

- Low latency response times
- Robust handling of diverse input scenarios

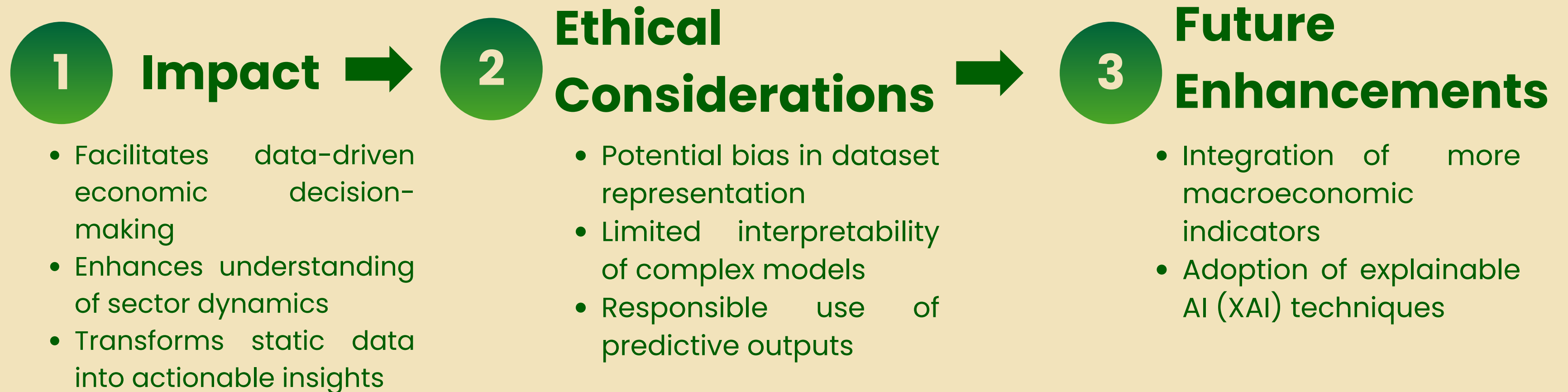
Application Domains

- Economic research and analysis
- Investment decision support
- Sector performance evaluation





REAL-WORLD IMPACT & ETHICAL CONSIDERATIONS





SYSTEM DEMONSTRATION

Demonstration Workflow

- Input or upload company data
- Execute the prediction model
- Visualize results through the dashboard

Outputs

- Growth predictions
- Cluster classification
- Interactive visual analytics

Demo video link:

https://drive.google.com/drive/folders/1D1Dof7iyBkeCpdZoWuEurGkT4evuKklo?usp=drive_link



CONCLUSION & KEY TAKEAWAYS

Key Achievements

- Successful development of a machine learning-based economic intelligence cluster system
- Integration of predictive modeling with full-stack web architecture
- Delivery of a real-time, interactive analytics dashboard

Key Insights

- Firm-level data provides deeper economic understanding than aggregate metrics
- Machine learning enhances predictive capability in complex datasets

Future Work

- Real-time data integration
- Enhanced model interpretability (XAI)
- Scalable and production-ready deployment



References

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Thank You

Leveraging advanced analytics and strategic insights to drive business growth and optimize marketing decision-making.

Q&A and Feedback

